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UNIT I

Chapter 2

Kingdom Animalia

Chapter Outline

2.1 Basis of Classification
2.2 Classification of Kingdom Animalia
2.3 Non Chordates (Invertebrata) upto Phyla level
2.4 Chordata upto Class level



March 20th is celebrated as World Sparrow day to conserve this endangered species, House sparrow (*Passer domesticus*)

Learning Objectives:

- *Justifies the need for classification.*
- *Understands the salient features of the animal phyla.*

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Kingdom Animalia comprises millions of animal species and studying them without a basic classification may lead to confusion. In addition to this, there are several new species of animals being constantly discovered. Classification is very essential for identification, naming and assigning a systematic position to the newly discovered species. Animal Kingdom is classified mainly based on the closely resembling characteristic features. Kingdom Animalia is characterised of eukaryotic, multicellular, heterotrophic organisms. They include about 35 phyla of which 11 are considered as major phyla. Almost 99 percent of animals are invertebrates or animals without backbone. The remaining represents vertebrates or animals with backbone. On the basis of the presence or absence of notochord (vertebral column), animals are also categorised into two major groups and they are non chordates and chordates.

2.1 Basis of classification

Multicellular organisms are structurally and functionally different but yet they possess certain common fundamental features such as the arrangement of cell layers, the levels of organisation, nature of coelom, the presence or absence of segmentation, notochord and the organisation of the organ system.

2.1.1. Levels of organisation

All members of Kingdom Animalia are metazoans (multicellular animals) and exhibit different patterns of cellular organisation. The cells of the metazoans